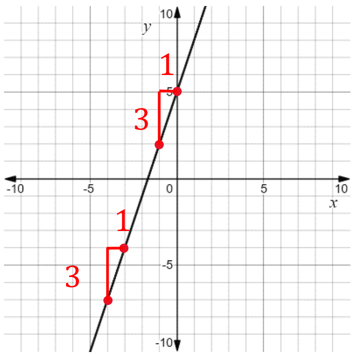
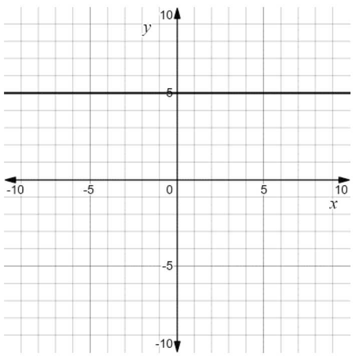
Algebra 1
Unit 3
Lesson 2 Practice

Name_Answer Key________
Date____________________
Period__________________

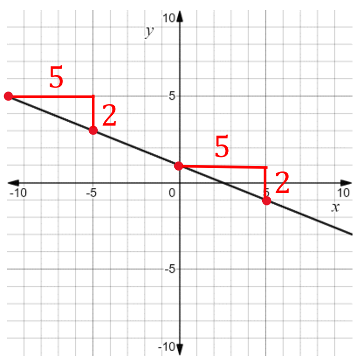
For questions 1 through 4, determine the slope of the line shown.

1. 

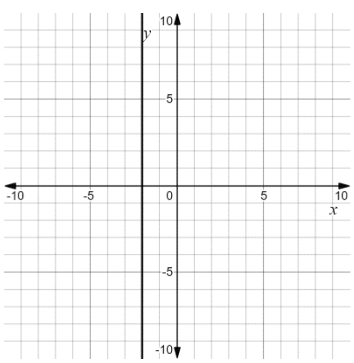
$m = \frac{3}{1} = 3$

2. 

$m = 0$

3. 

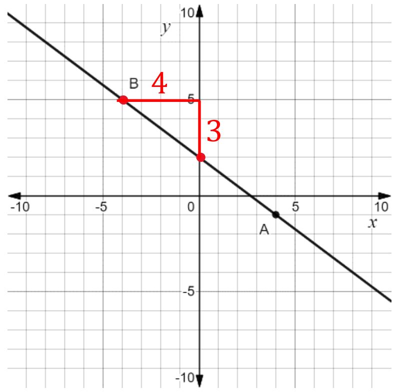
$m = -\frac{2}{5}$

4. 

$m =$ undefined

For questions 5 through 8, the graph of a line with points A and B labeled is shown.

$m = -\frac{3}{4}$



point $B = (-4,5)$
point $A = (4,-1)$

5. Write the equation of the line in point-slope form using point A.

$y + 1 = -\frac{3}{4}(x - 4)$

6. Write the equation of the line in point-slope form using point B.

$y - 5 = -\frac{3}{4}(x + 4)$

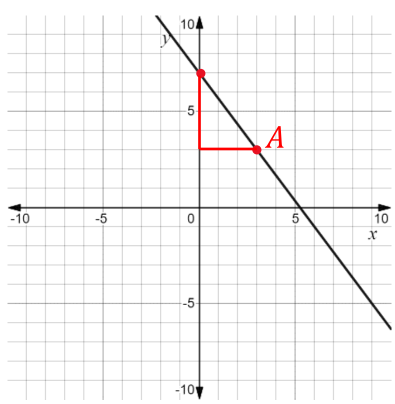
7. Rewrite the equation in point-slope form from point A in slope-intercept form.

$y + 1 = -\frac{3}{4}x + 3 \quad \rightarrow \quad y = -\frac{3}{4}x + 2$

8. Rewrite the equation in point-slope form from point B in standard form.

$4y - 20 = -3(x + 4) \quad \rightarrow \quad 4y - 20 = -3x - 12 \quad \rightarrow \quad 3x + 4y = 8$

For questions 9 and 10, the graph of a line is shown.



$m = -\frac{4}{3}$
point $A = (3,3)$

9. Write the equation of the line in point-slope form.

$y - 3 = -\frac{4}{3}(x - 3)$

10. Rewrite the equation from question 9 in standard form.

$y - 3 = -\frac{4}{3}x + 4 \quad \rightarrow \quad \frac{4}{3}x + y - 3 = 4 \quad \rightarrow \quad \frac{4}{3}x + y = 7$ or $4x + 3y = 21$